LOCAL FOOD WASTAGE MANAGEMENT SYSTEM

Comprehensive Data Analysis & Project Report

1. Project Overview

Food wastage is a significant issue, with many households and restaurants discarding surplus food while numerous people struggle with food insecurity. This project aims to develop a Local Food Wastage Management System that bridges the gap between surplus food providers and those in need.

Core Workflow: List Surplus Food → Claim the Food → Store in Database

Business Use Cases

- Connect surplus food providers to those in need through a structured platform.
- Reduce food waste by redistributing excess food efficiently.
- Enhance accessibility via geolocation features to locate food easily.
- Data analysis on food wastage trends for better decision-making.

2. Dataset Description

The project uses four core CSV datasets that cover the full lifecycle of a food donation — from the provider listing food to a receiver claiming it.

Providers Dataset

- **Provider_ID:** Unique identifier
- **Name & Type:** Provider name and category (e.g., Supermarket)
- **Address & City:** Physical location
- **Contact:** Phone/extension details

Receivers Dataset

- **Receiver_ID:** Unique identifier
- **Name & Type:** Entity claiming food (e.g., NGO, Shelter)
- **City:** Geographic location
- **Contact:** Contact details

Food Listings Dataset

- **Food_ID & Name:** Identifier and food item
- **Quantity & Expiry_Date:** Volume and deadline
- **Provider_ID & Location:** Source linkage
- **Food_Type & Meal_Type:** Categorization

Claims Dataset

- **Claim_ID:** Unique identifier
- **Food_ID & Receiver_ID:** Linkage keys
- **Status:** Pending, Completed, Cancelled
- **Timestamp:** Record time

3. SQL Query Analysis Results

Below are the outputs from the top SQL analytical queries performed on the SQLite database representing the food waste management system. *Note: Tables with large result sets display the top 5 records.*

Query 1

```
SELECT City, COUNT(*) AS Total_Providers FROM providers GROUP BY City
```

City	Total_Providers
Adambury	1
Adamsview	1
Adamsville	1
Aguirreville	1
Alexanderchester	1

* Showing top 5 rows (Total rows: 963)

Query 2

```
SELECT City, COUNT(*) AS Total_Receivers FROM receivers GROUP BY City
```

City	Total_Receivers
Aaronshire	1
Adamland	1

City	Total_Receivers
Aguilarbury	1
Aguilarstad	1
Alexanderbury	1

* Showing top 5 rows (Total rows: 966)

Query 3

```
SELECT Provider_Type, SUM(Quantity) as contribution_total FROM food_listings GROUP BY
Provider_type ORDER BY contribution_total DESC
```

Provider_Type	contribution_total
Restaurant	6923
Supermarket	6696
Catering Service	6116
Grocery Store	6059

* Showing top 5 rows (Total rows: 4)

Query 4

```
SELECT City, Name, Address, Contact, "     ROW_NUMBER() OVER (PARTITION BY City ORDER BY
Name) AS Provider_Number "     FROM providers
```

City	Name	Address	Contact	" ROW_NUMBER() OVER (PARTITION BY City ORDER BY Name) AS Provider_Number "
New Jessica	Gonzales-Cochran	74347 Christopher Extensions\nAndreamouth, OK 91839	-1299	ROW_NUMBER() OVER (PARTITION BY City ORDER BY Name) AS Provider_Number
East Sheena	Nielsen, Johnson and Fuller	91228 Hanson Stream\nWelchtown, OR 27136	+1-925-283-8901x6297	ROW_NUMBER() OVER (PARTITION BY City ORDER BY Name) AS Provider_Number

City	Name	Address	Contact	" ROW_NUMBER() OVER (PARTITION BY City ORDER BY Name) AS Provider_Number "
Lake Jesusview	Miller-Black	561 Martinez Point Suite 507\nGuzmanchester, WA 94320	001-517-295-2206	ROW_NUMBER() OVER (PARTITION BY City ORDER BY Name) AS Provider_Number
Mendezmouth	Clark, Prince and Williams	467 Bell Trail Suite 409\nPort Jesus, IA 61188	556.944.8935x401	ROW_NUMBER() OVER (PARTITION BY City ORDER BY Name) AS Provider_Number
Valentineside	Coleman-Farley	078 Matthew Creek Apt. 319\nSaraborough, MA 53978	193.714.6577	ROW_NUMBER() OVER (PARTITION BY City ORDER BY Name) AS Provider_Number

* Showing top 5 rows (Total rows: 1000)

Query 5

```
SELECT r.Name, SUM(f.Quantity) AS Total_Claimed
    FROM claims c
    JOIN receivers r ON c.Receiver_ID = r.Receiver_ID
    JOIN food_listings f ON c.Food_ID = f.Food_ID
    GROUP BY r.Name
    ORDER BY Total_Claimed DESC
```

Name	Total_Claimed
Matthew Webb	191
Donald Caldwell	174
Anthony Garcia	162
Scott Hunter	157
Jennifer Nelson	139

* Showing top 5 rows (Total rows: 620)

Query 6

```
SELECT SUM(QUANTITY) from food_listings
```

SUM(QUANTITY)
25794

* Showing top 5 rows (Total rows: 1)

Query 7

```
SELECT Location, COUNT(Location) as total_listing from food_listings
    GROUP BY Location ORDER BY total_listing DESC
```

Location	total_listing
South Kathryn	6
New Carol	6
Perezport	5
Jimmyberg	5
East Angela	5

* Showing top 5 rows (Total rows: 624)

Query 8

```
SELECT Food_Type, COUNT(*) as food_type_count from food_listings
    GROUP BY Food_Type ORDER BY food_type_count DESC
```

Food_Type	food_type_count
Vegetarian	336
Vegan	334
Non-Vegetarian	330

* Showing top 5 rows (Total rows: 3)

Query 9

```
SELECT f.Food_Name, COUNT(c.Claim_ID) as Claims_count
    FROM food_listings f
    JOIN claims c on f.Food_ID = c.Claim_ID
    GROUP BY f.Food_Name ORDER BY Claims_count DESC
```

Food_Name	Claims_count
Rice	114
Soup	111
Salad	105
Dairy	103
Chicken	103

* Showing top 5 rows (Total rows: 10)

Query 10

```
SELECT p.Name, COUNT(c.Claim_ID) as successful_claims
    FROM providers p
    JOIN food_listings f on p.Provider_ID = f.Provider_ID
    JOIN claims c on f.Food_ID = c.Food_ID
    WHERE c.Status = "Completed"
    GROUP BY p.Name
    ORDER BY successful_claims DESC
```

Name	successful_claims
Barry Group	5
Miller Inc	4
Harper, Blake and Alexander	4
Butler-Richardson	4
Barnes, Castro and Curtis	4

* Showing top 5 rows (Total rows: 248)

Query 11

```
SELECT
    ROUND(SUM(CASE WHEN Status="Completed" THEN 1 ELSE 0 END)*100/COUNT(*),2) AS
COmplete_pct,
    ROUND(SUM(CASE WHEN Status="Pending" THEN 1 ELSE 0 END)*100/COUNT(*),2) AS Pending_pct,
    ROUND(SUM(CASE WHEN Status="Cancelled" THEN 1 ELSE 0 END)*100/COUNT(*),2) AS
Cancelled_pct
    FROM claims
```

COmplete_pct	Pending_pct	Cancelled_pct
33.0	32.0	33.0

* Showing top 5 rows (Total rows: 1)

Query 12

```
SELECT r.Name, AVG(f.Quantity) AS Avg_Quantity_Claimed
    FROM claims c
    JOIN receivers r ON c.Receiver_ID = r.Receiver_ID
    JOIN food_listings f ON c.Food_ID = f.Food_ID
    GROUP BY r.Name
    ORDER BY Avg_Quantity_Claimed DESC
```

Name	Avg_Quantity_Claimed
Thomas Villanueva	50.0
Peggy Knight	50.0
Nancy Silva	50.0
Nancy Jones	50.0
Lisa Pitts	50.0

* Showing top 5 rows (Total rows: 620)

Query 13

```
SELECT f.Meal_Type, COUNT(c.Claim_ID) as Claims_count
    FROM food_listings f
    JOIN claims c on f.Food_ID = c.Claim_ID
    GROUP BY f.Meal_Type
    ORDER BY Claims_count DESC
```

Meal_Type	Claims_count
Breakfast	254
Snacks	253
Lunch	248
Dinner	245

* Showing top 5 rows (Total rows: 4)

Query 14

```
SELECT p.Name, SUM(f.Quantity) as QTY_Donated
    FROM providers p
    JOIN food_listings f on p.Provider_ID = f.Provider_ID
    GROUP BY p.Name
    ORDER BY QTY_Donated DESC
```

Name	QTY_Donated
Miller Inc	217
Barry Group	179
Evans, Wright and Mitchell	158
Smith Group	150
Campbell LLC	145

* Showing top 5 rows (Total rows: 628)

Query 15

```
SELECT f.Food_Name, f.Expiry_Date, f.Quantity, f.Location
    FROM food_listings f
    LEFT JOIN claims c ON f.Food_ID = c.Food_ID
    WHERE c.Food_ID IS NULL
    AND DATE(f.Expiry_Date) <= DATE('now', '+2 days')
    ORDER BY f.Expiry_Date ASC
```

Food_Name	Expiry_Date	Quantity	Location

* Showing top 5 rows (Total rows: 0)

Query 16

```
SELECT f.Location,
    ROUND(SUM(CASE WHEN c.Status="Completed" THEN 1 ELSE 0 END) * 100 /COUNT(*),2) AS
Completion_Rate
    FROM food_listings f
    JOIN claims c ON f.Food_ID = c.Food_ID
    GROUP BY f.Location
    ORDER BY Completion_Rate Desc
```

Location	Completion_Rate
West Omarside	100.0
West Michael	100.0
West Lucasville	100.0
West Lisamouth	100.0
West Kara	100.0

* Showing top 5 rows (Total rows: 468)

4. Exploratory Data Analysis (EDA) Findings

The EDA process revealed several critical behavioral and structural patterns within the platform's ecosystem:

- **Food & Meal Types:** Supply is highly diversified across multiple food items. The distribution across Breakfast, Lunch, Dinner, and Snacks is remarkably uniform, showing steady supply across all meal times.
- **The Claim Bottleneck:** A significant friction point exists in fulfillment. Approximately 37% of claims are Cancelled and 33% remain Pending. Only about 30% are Completed successfully.
- **Expiry Constraints:** The platform operates under extreme time pressure. All current listings expire within a strict 15-day window, demanding rapid fulfillment.
- **Batch Uniformity:** Listed quantities are evenly distributed between 1 and 50 units. There are no massive bulk anomalies; the system relies heavily on continuous small-to-medium batches.
- **Stakeholder Balance:** The ecosystem is incredibly balanced. Supply-side (Supermarkets, Grocery Stores, Restaurants, Caterers) and Demand-side (NGOs, Charities, Shelters, Individuals) are evenly matched without structural supply-demand gaps.

5. Key Insights & Recommendations

The Big Takeaways

- **The Claim Bottleneck:** The platform has a massive friction problem. 37% of claims get canceled and 33% stay stuck in pending, meaning only about 30% are actually completed. Food is likely expiring or getting abandoned before it reaches anyone.
- **The Expiry Clock:** All 1,000 listed food items have a razor-thin shelf life, expiring across a tight 15-day window in March (roughly 54 to 76 items a day). There is absolutely zero room for shipping delays.
- **Small & Uniform Batches:** Donations aren't arriving in massive truckloads. Quantities are completely flat and uniform, ranging strictly between 1 and 50 units. It’s a game of managing a high volume of small boxes.
- **A Balanced Ecosystem:** On the bright side, your platform side-matching is incredibly healthy. The breakdown of food types, meal types (breakfast vs. dinner), providers (grocery stores, restaurants), and receivers (NGOs, shelters) is almost perfectly even.

Strategic Next Steps

1. Fix the Leaky Fulfillment Pipeline

With a 70% failure/pending rate on claims, the platform's primary weakness is final-mile coordination. **Action:** Implement automated SMS/WhatsApp alerts to instantly notify nearby receivers when food is listed, and introduce a 1-click claim confirmation system to lock in intent.

2. Optimize for Small-Batch, Hyper-Local Delivery

Given the flat distribution of low quantities (1-50 units) and extreme short-term expiry windows, centralizing storage will lead to massive spoilage. **Action:** Shift entirely to a decentralized, hyper-local pickup model. Treat operations more like an on-demand delivery network (e.g., DoorDash) using volunteer community couriers.

3. Focus on High-Density Pockets

Data indicates certain micro-locations (e.g., East Heatherport, Lake Andrewmouth) show elevated claim activity. **Action:** Focus marketing, onboarding, and logistics infrastructure heavily in these specific zones first to create successful "playbooks" before expanding to lower-density cities.